

# Volume V · Chapter 23 — Robotics Fundamentals & Simulation (ROS 2, Isaac Sim) · Theory

Source: split from `med-tech-curriculum-V1.md` (V1). From this split onward this chapter is maintained **independently** here. See `Volume-V-Chapter-23-context.md` for the AI interaction log.

**Prerequisites:** Ch 4; general programming maturity.

**Learning outcomes — the student can:** explain robot kinematics/control basics; use ROS 2 concepts (nodes, topics, services, actions); run Gazebo/Isaac Sim; command a simulated arm.

**Topics (theory) — 10 h:**

| #    | Topic                                                                     | h |
|------|---------------------------------------------------------------------------|---|
| 23.1 | Robotics basics:<br>kinematics, DoF,<br>coordinate frames                 | 2 |
| 23.2 | Control loops & real-<br>time constraints                                 | 2 |
| 23.3 | ROS 2 architecture:<br>nodes, topics, services,<br>actions                | 3 |
| 23.4 | Simulation: Gazebo &<br>NVIDIA Isaac Sim                                  | 2 |
| 23.5 | Medical robotics<br>landscape (surgical,<br>assistive, lab<br>automation) | 1 |

**Datasets/tools:** ROS 2, Gazebo, Isaac Sim; GPU workstation. **Assessment:** simulation task deliverable (**60%**); ROS 2 quiz (**20%**); writeup (**20%**). **Key decisions:** simulator choice; **sim vs. hardware**; real-time constraints. **References:** plan §7; §13 → *Robotics & embodied AI*. **Hours:** Theory **10** + Lab **16** = **26**.